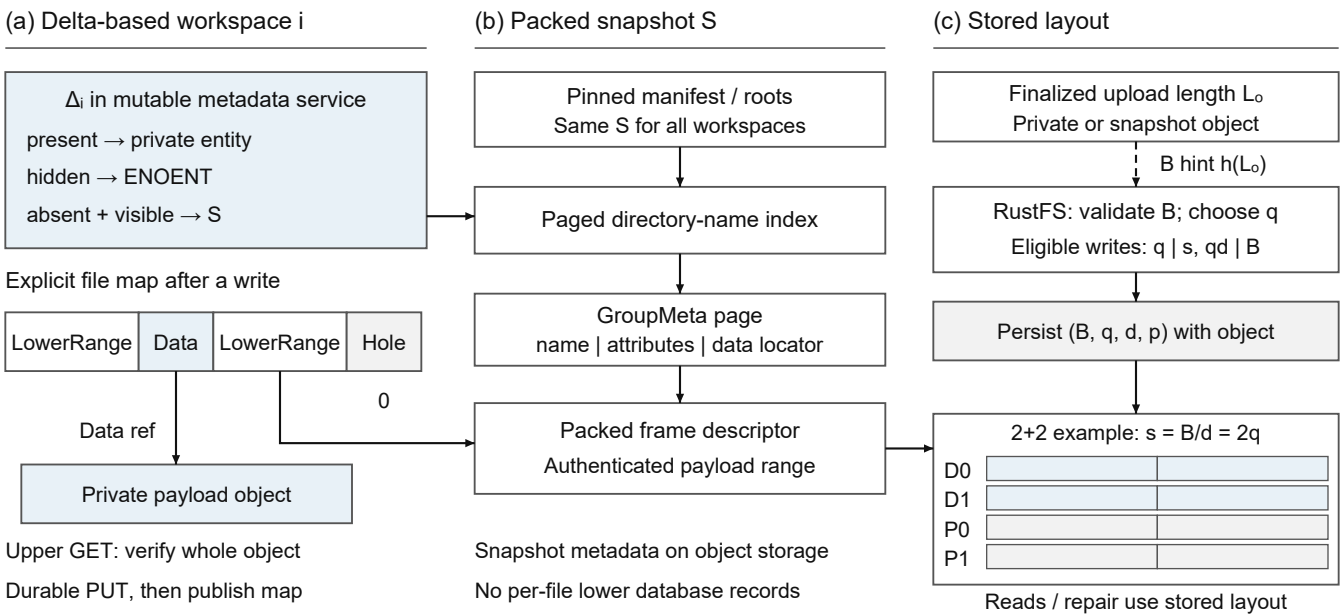
(a) Delta-based workspace i
(b) Packed snapshot S
(c) Stored layout
Δi in mutable metadata service
present → private entity
hidden → ENOENT
absent + visible → S
Pinned manifest / roots
Same S for all workspaces
Finalized upload length Lo
Private or snapshot object
B hint h(Lo)
Paged directory-name index
RustFS: validate B; choose q
Eligible writes: q | s, qd | B
Explicit file map after a write
LowerRange
Data
LowerRange
Hole
0
GroupMeta page
name | attributes | data locator
Persist (B, q, d, p) with object
Data ref
Packed frame descriptor
Authenticated payload range
2+2 example: s = B/d = 2q
D0
D1
P0
P1
Private payload object
Upper GET: verify whole object
Durable PUT, then publish map
Snapshot metadata on object storage
No per-file lower database records
Reads / repair use stored layout